Homework Assignment: Jira Integration

You already have the existing application with *ingestion, retrieval, MongoDB, frontend, and test case generation* functionality.

For this assignment, extend the existing application by integrating *Jira*.

Requirements

* [ ] Integrate *Jira* with the existing application.

* [ ] Fetch the *user story from Jira*.

* [ ] Ingest the fetched user story into *MongoDB*.

* [ ] Retrieve the existing/relevant context from MongoDB.

* [ ] Use the Jira user story along with the retrieved context to *generate test cases* using the existing functionality.

* [ ] Display the generated test cases in the frontend.

Expected Flow

Jira → Fetch User Story → Ingest into MongoDB → Retrieve Existing Context → Generate Test Cases → Display Results

Note: You do not need to rebuild the existing ingestion, retrieval, or test case generation functionality. Focus on the *Jira integration and connecting it with the existing flow*.

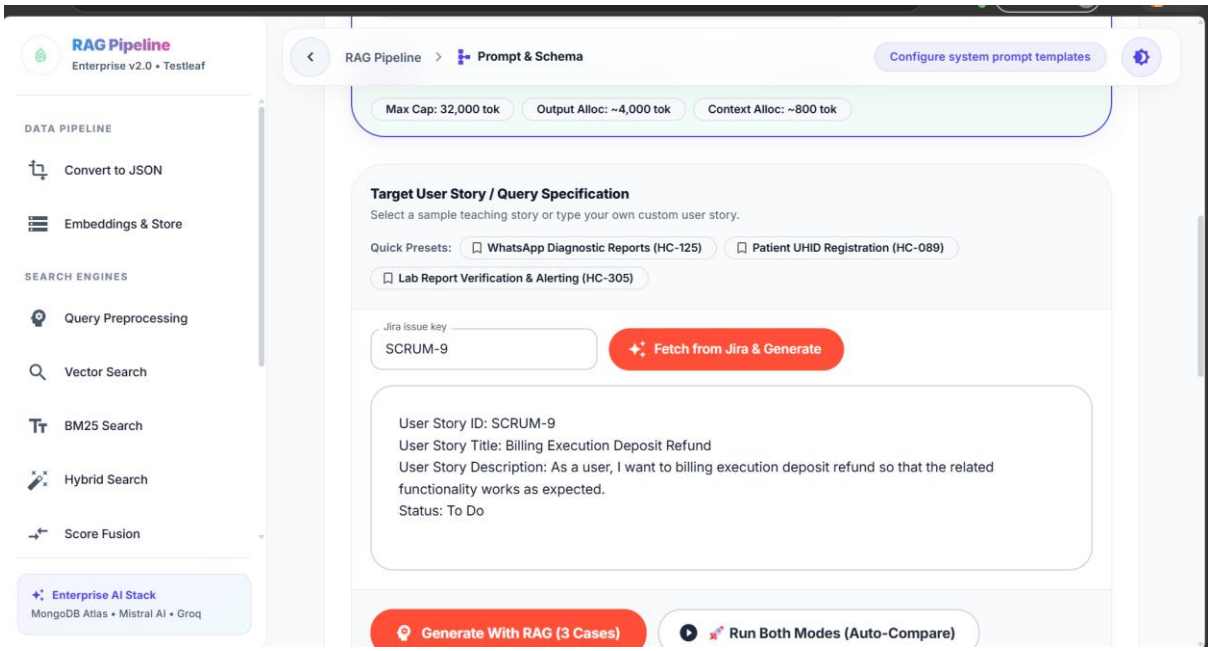
JIRA Integration- Implementation

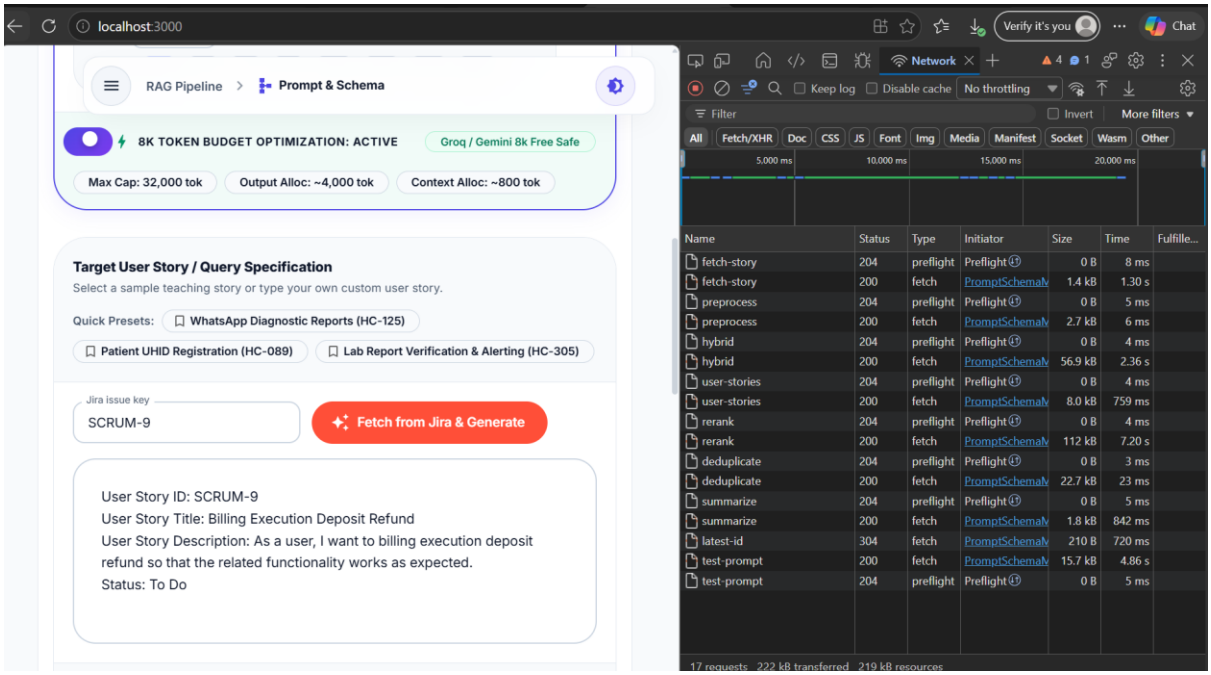
Added the JIRA integration details in .env file

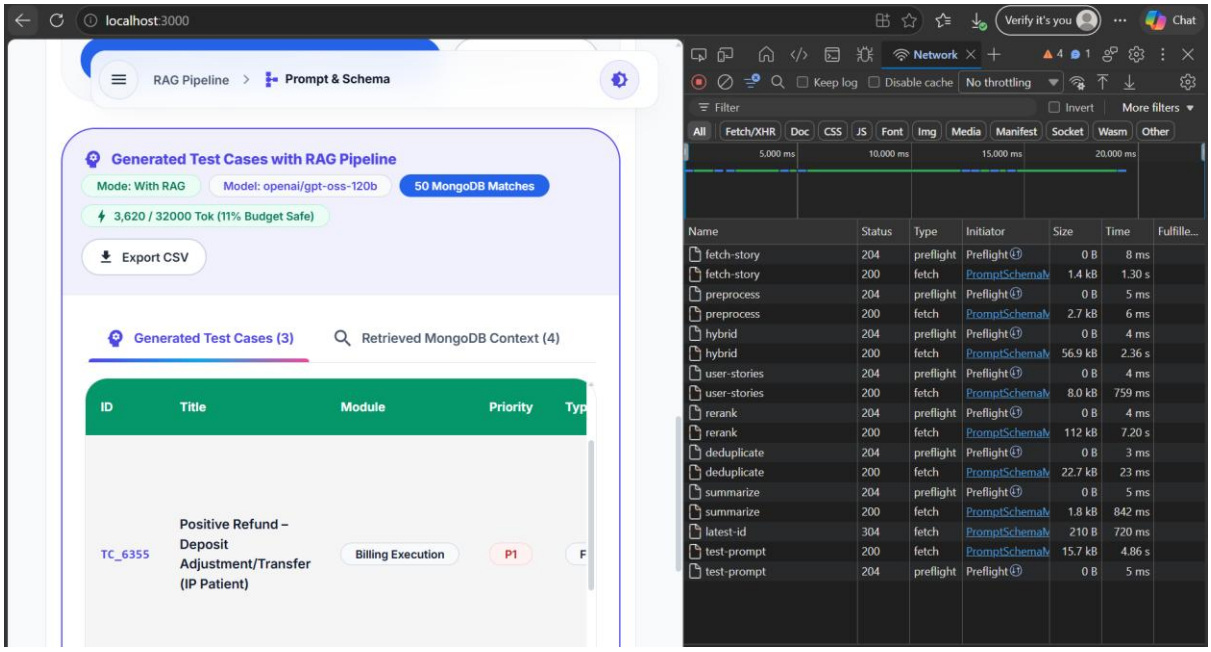
user-stories-bm25-index.json | package.json | .env | us-310.txt

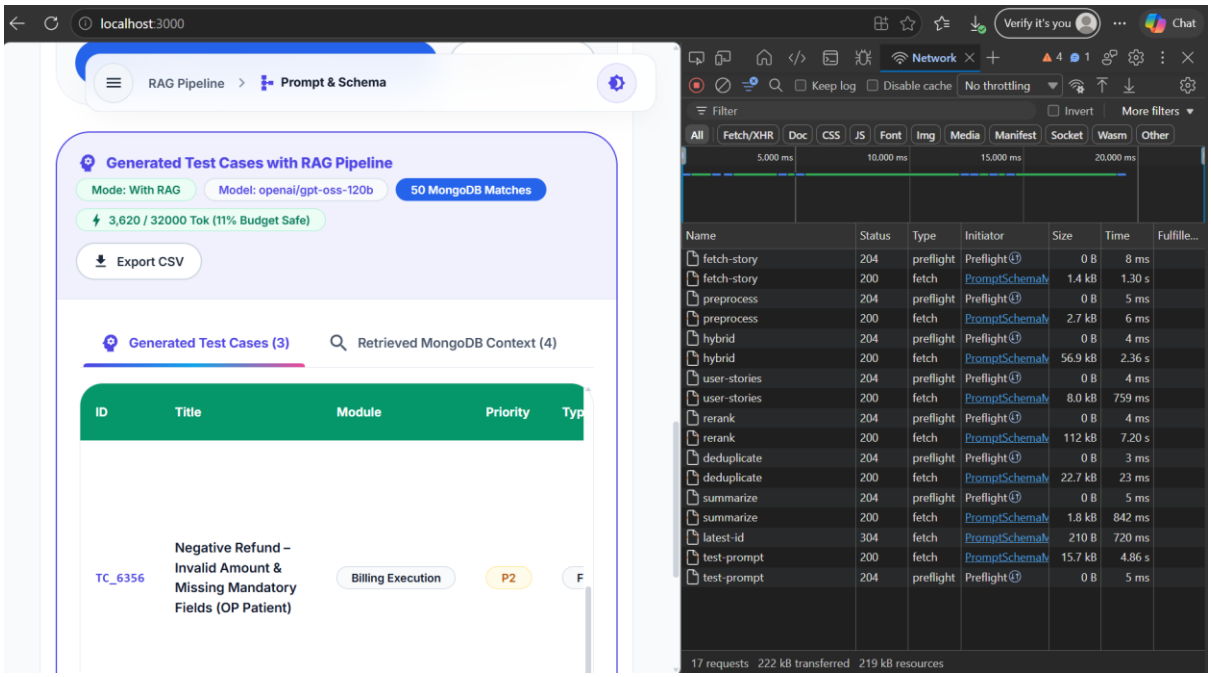
.env

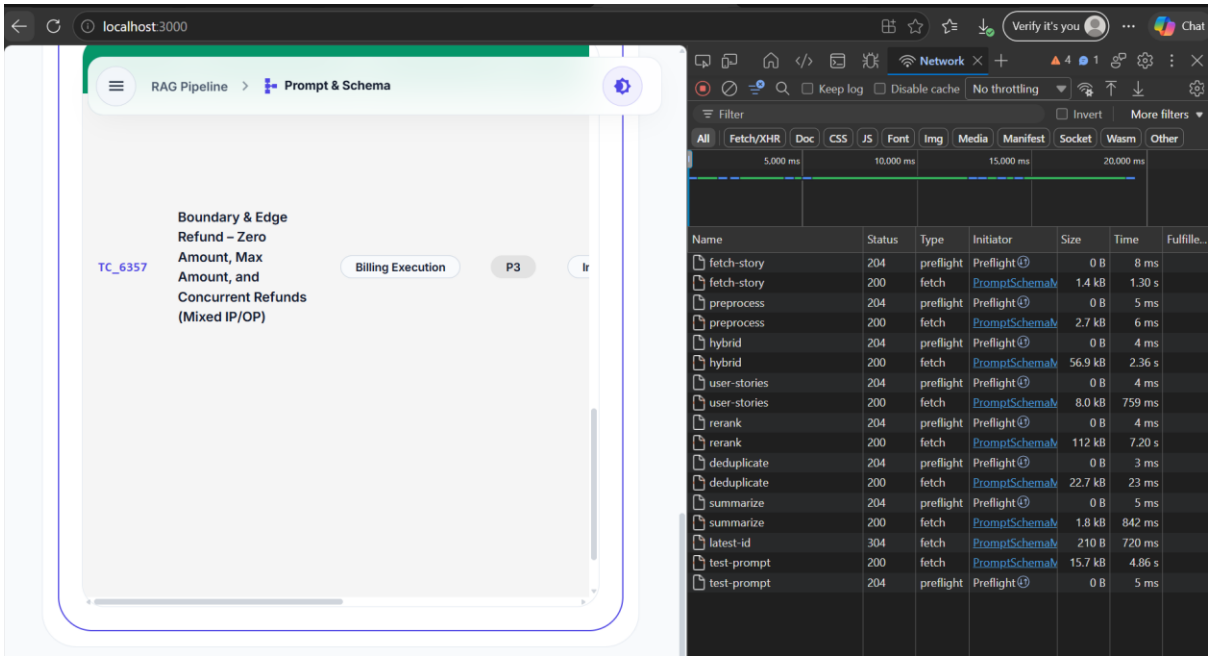
```
49
50  # ======================== API Keys Vault ========================
51  GROQ_API_KEY=gsk_MqkI3LOjeCGdUFfUeoVKWGdyb3FYUJ8YmBbJnpeRNG40hdcg
52  GEMINI_API_KEY="AQ.Ab8RN6LPnE1NfoiUOzDoQ1sg4sHSTLG9h8oBd1cbbJuo8k
53  OPENAI_API_KEY=""
54  ANTHROPIC_API_KEY=""
55  MISTRAL_API_KEY="mstrl_RXAzP2fR9rZkOHeK4AdGChZPbdL0r5na_0uzeZ0"
56  NVIDIA_API_KEY="nvapi-9FGjeAdRpyuvVFAwYC7vkcbAFU_se3tlTVu1PBq_1-A
57  NVIDIA_BASE_URL="https://integrate.api.nvidia.com/v1/"
58
59  # Jira Cloud Integration
60  JIRA_BASE_URL="https://archanam40.atlassian.net"
61  JIRA_EMAIL="archanam40@gmail.com"
62  JIRA_API_TOKEN="ATATT3xFfGF0ZtLpL-43FGIeJTSVAxvFvwq-kVhft_h7d0Nmc
63  JIRA_PROJECT_KEY="GENAI PILOTS"
64
```




localhost:3000
Verify it's you
Chat
RAG Pipeline
Prompt & Schema
8K TOKEN BUDGET OPTIMIZATION: ACTIVE
Groq / Gemini 8k Free Safe
Max Cap: 32,000 tok
Output Alloc: ~4,000 tok
Context Alloc: ~800 tok
Target User Story / Query Specification
Select a sample teaching story or type your own custom user story.
Quick Presets:
WhatsApp Diagnostic Reports (HC-125)
Patient UHID Registration (HC-089)
Lab Report Verification & Alerting (HC-305)
Jira issue key
SCRUM-9
Fetch from Jira & Generate
User Story ID: SCRUM-9
User Story Title: Billing Execution Deposit Refund
User Story Description: As a user, I want to billing execution deposit refund so that the related functionality works as expected.
Status: To Do
Network
Keep log
Disable cache
No throttling
Filter
Invert
More filters
All
Fetch/XHR
Doc
CSS
JS
Font
Img
Media
Manifest
Socket
Wasm
Other
5,000 ms
10,000 ms
15,000 ms
20,000 ms
Name
Status
Type
Initiator
Size
Time
Fulfille...
fetch-story 204 preflight Preflight 0 B 8 ms
fetch-story 200 fetch PromptSchemaM 1.4 kB 1.30 s
preprocess 204 preflight Preflight 0 B 5 ms
preprocess 200 fetch PromptSchemaM 2.7 kB 6 ms
hybrid 204 preflight Preflight 0 B 4 ms
hybrid 200 fetch PromptSchemaM 56.9 kB 2.36 s
user-stories 204 preflight Preflight 0 B 4 ms
user-stories 200 fetch PromptSchemaM 8.0 kB 759 ms
rerank 204 preflight Preflight 0 B 4 ms
rerank 200 fetch PromptSchemaM 112 kB 7.20 s
deduplicate 204 preflight Preflight 0 B 3 ms
deduplicate 200 fetch PromptSchemaM 22.7 kB 23 ms
summarize 204 preflight Preflight 0 B 5 ms
summarize 200 fetch PromptSchemaM 1.8 kB 842 ms
latest-id 304 fetch PromptSchemaM 210 B 720 ms
test-prompt 200 fetch PromptSchemaM 15.7 kB 4.86 s
test-prompt 204 preflight Preflight 0 B 5 ms
17 requests 222 kB transferred 219 kB resources


localhost:3000
Verify it's you
Chat
RAG Pipeline
Prompt & Schema
Generated Test Cases with RAG Pipeline
Mode: With RAG
Model: openai/gpt-oss-120b
50 MongoDB Matches
3,620 / 32000 Tok (11% Budget Safe)
Export CSV
Generated Test Cases (3)
Retrieved MongoDB Context (4)
ID
Title
Module
Priority
Typ
TC_6355
Positive Refund – Deposit Adjustment/Transfer (IP Patient)
Billing Execution
P1
F
Network
Keep log
Disable cache
No throttling
Filter
Invert
More filters
All
Fetch/XHR
Doc
CSS
JS
Font
Img
Media
Manifest
Socket
Wasm
Other
5,000 ms
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Mode: With RAG
Model: openai/gpt-oss-120b
50 MongoDB Matches
3,620 / 32000 Tok (11% Budget Safe)
Export CSV
Generated Test Cases (3)
Retrieved MongoDB Context (4)
ID
Title
Module
Priority
Typ
TC_6356
Negative Refund – Invalid Amount & Missing Mandatory Fields (OP Patient)
Billing Execution
P2
F
Network
Keep log
Disable cache
No throttling
Filter
Invert
More filters
All
Fetch/XHR
Doc
CSS
JS
Font
Img
Media
Manifest
Socket
Wasm
Other
5,000 ms
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15,000 ms
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Logs

[0] ## INSTRUCTION

[0] Generate high-quality test cases for the user story using retrieved te..."

[0] ==========

[0] ✅ [LLM COMPLETED] Provider: [GROQ] | Time: 7132ms | Tokens: 4833

[0]

[0] ✅ LLM response received from [groq] model [openai/gpt-oss-120b]

[0] 🧠 [VECTOR EMBEDDING] Provider: [MISTRAL] | Model: [mistral-embed]

[0] 🔀 Hybrid Search request: "User Story ID: SCRUM-9

[0] User Story Title: Billing Execution Deposit Refund

[0] User Story Description: As a user, I want to billing execution deposit refund so that the related functionality works as expected.

[0] Status: To Do"

[0] BM25 Weight: 0.4, Vector Weight: 0.6

[0] 🔤 Running BM25 search with weighted field boosting...

[0] ✅ BM25 search returned 200 candidates in 146ms

[0] 🧠 Running vector search...

[0] 🧠 [VECTOR EMBEDDING] Provider: [MISTRAL] | Model: [mistral-embed]

[0] ✅ Vector search returned 200 candidates in 1902ms

[0] 🔀 Combining results with score fusion...

[0] ✅ Hybrid Search complete: 50 distinct results in 2054ms

[0]

[0] 📖 User Stories Hybrid Search

[0] Query: "User Story ID: SCRUM-9

[0] User Story Title: Billing Execution Deposit Refund

[0] User Story Description: As a user, I want to billing execution deposit refund so that the related functionality works as expected.

[0] Status: To Do"

[0] Collection: userstories

[0] Vector Index: vector_index_userstories

[0] BM25 Index: bm25_user_stories

[0] 🧠 Running vector search on user_stories...

[0] 🧠 [VECTOR EMBEDDING] Provider: [MISTRAL] | Model: [mistral-embed]

[0] ✅ Vector search: 30 results in 374ms

[0] 🔤 Running BM25 search on user_stories...

[0] ✅ BM25 search: 0 results in 26ms

[0] ✅ User Stories search complete: 10 results in 752ms

[0]

[0] 🤖 SEMANTIC RERANKING MODE

[0] Query: "User Story ID: SCRUM-9

[0] User Story Title: Billing Execution Deposit Refund

[0] User Story Description: As a user, I want to billing execution deposit refund so that the related functionality works as expected.

[0] Status: To Do"

[0] Limit: 10

[0] Provider: [GROQ] | Model: [openai/gpt-oss-120b]

[0] Filters: {}

[0] 📊 Fetching candidates from BM25 and Vector search (target: 50)...

[0] 🧠 [VECTOR EMBEDDING] Provider: [MISTRAL] | Model: [mistral-embed]

[0] ✅ Retrieved & fused 47 candidates from RAG (original ranks 1..47)

[0] 🤖 Sending 47 candidates to LLM for semantic reranking...

[0]

[0] ==

[0] 🔄 [SEMANTIC RERANKING]

[0] 🏷️ Provider: [GROQ] | Model: [openai/gpt-oss-120b]

[0] 🔍 Query: "User Story ID: SCRUM-9

[0] User Story Title: Billing Execution Deposit Refund

[0] User Story Description: As a user, I want to billing execution deposit refund so that the related functionality works as expected.

[0] Status: To Do" | Candidate Pool: 47 | Scoring: 47

[0] ==

[0]

[0] ==

[0] 🤖 [LLM INVOCATION] Provider: [GROQ] | Model: [openai/gpt-oss-120b]

[0] 📩 Prompt: "You are an expert search relevance evaluator. You must evaluate and assign a relevance score (0-100) to EVERY SINGLE doc..."

[0] ==

[0]

[0] ==

[0] 🤖 [LLM INVOCATION] Provider: [GROQ] | Model: [openai/gpt-oss-120b]

[0] 📩 Prompt: "You are an expert search relevance evaluator. You must evaluate and assign a relevance score (0-100) to EVERY SINGLE doc..."

[0] ==

[0] ✅ [LLM COMPLETED] Provider: [GROQ] | Time: 3439ms | Tokens: 2939

[0]

[0] ✅ [LLM COMPLETED] Provider: [GROQ] | Time: 4365ms | Tokens: 3330

[0]

[0] ✅ [SEMANTIC RERANKING COMPLETED] Top-10 documents scored & re-ordered successfully (Scored: 47/47)

[0]

[0] ✅ Semantic reranking complete in 4371ms

[0] 🖌️ Deduplicating 10 items with threshold 0.95 (raw input: 0.95)

[0]

[0] ==========

[0] 📝 [SUMMARIZATION & DEDUPLICATION]

[0] 🏷️ Provider: [GROQ] | Model: [qwen/qwen3.8-27b]

[0] 🔍 Query: "test cases" | Style: concise | Input Documents: 4

[0] ==========

[0]

[0] ==========

[0] 🤖 [LLM INVOCATION] Provider: [GROQ] | Model: [qwen/qwen3.8-27b]

[0] 📩 Prompt: "You are an expert QA and test case search analyst.

[0]

[0] Query: "test cases"

[0]

[0] Search Results:

[0] 1. ID: TC_3410 | Title: Billing..."

[0] ==========

[0] ✅ [LLM COMPLETED] Provider: [GROQ] | Time: 830ms | Tokens: 707

[0]

[0] ✅ [SUMMARIZATION COMPLETED] Summary generated (1177 chars)

[0]

[0] ✅ Summarization complete

[0] 🤖 Running LLM prompt generation...

[0]

[0] ==

[0] 🤖 [LLM INVOCATION] Provider: [GROQ] | Model: [openai/gpt-oss-120b]

[0] 📩 Prompt: "# HEALTHCARE TEST CASE GENERATION

[0]

[0] ## INSTRUCTION

[0] Generate high-quality test cases for the user story using retrieved te..."

[0] ==

[0] ✅ [LLM COMPLETED] Provider: [GROQ] | Time: 4844ms | Tokens: 3620

[0]

[0] ✅ LLM response received from [groq] model [openai/gpt-oss-120b]